Chest X-Ray Pneumonia Classification Using Deep Learning and Ensemble Approaches

Fazly Fardin Chowdhury
2322480642

Sudipto Roy
2222756042

Syed Mohaiminul Islam
2322429042

Abstract

This report presents a comprehensive, end-to-end comparative study of machine learning and deep learning architectures applied to automated pneumonia detection from chest X-ray images. The publicly available Chest X-Ray Images (Pneumonia) dataset by Paul Mooney (Kaggle) is used, comprising 5,856 labeled JPEG images across two classes—Normal and Pneumonia—split into training (5,216), validation (16), and test (624) sets. Nine distinct modeling strategies are systematically implemented, trained, and evaluated: (1) a traditional Random Forest classifier using Histogram of Oriented Gradients (HOG) feature extraction, (2) a custom Convolutional Neural Network (CNN) built with TensorFlow/Keras, (3) an Autoencoder-based feature extractor coupled with a downstream classifier, (4) Transfer Learning using MobileNetV2, (5) Transfer Learning using ResNet50 with frozen and fine-tuned configurations, (6) EfficientNetB0 with frozen backbone, (7) VGG19 fine-tuning with progressive unfreeze in two augmentation variants, (8) Swin Transformer in a 4-variant ablation study, (9) DeiT in three setups, and (10) a four-model ensemble of PyTorch-trained ResNet50, VGG19-BN, Swin-Base, and DeiT-Base, optimized via Optuna Bayesian search with GradCAM explainability. Results show MobileNetV2 achieving the highest single-model test accuracy of 96.46%, while the untuned soft-voting ensemble achieves 91.19% accuracy with 99.7% recall on Pneumonia—the highest sensitivity in the study and the most clinically appropriate configuration for screening.

Keywords: *pneumonia detection, chest X-ray, deep learning, transfer learning, vision transformers, ensemble learning, Bayesian hyperparameter optimization, GradCAM*

I. Introduction

Pneumonia is an acute respiratory infection that inflames the air sacs in one or both lungs. It is responsible for approximately 14% of all deaths of children under five years of age worldwide and remains a leading cause of hospitalization among elderly populations. Early and accurate diagnosis is critical for timely antibiotic or antiviral treatment; however, conventional interpretation of chest radiographs requires trained radiologists and is prone to inter-observer variability, particularly in resource-constrained settings.

The rapid advancement of deep learning—particularly CNNs and, more recently, Vision Transformers—has catalyzed a new era of computer-aided diagnosis. These models can identify pathological patterns in medical images with accuracy rivaling clinical experts, and can operate at scale without fatigue. Automated pneumonia screening from chest X-rays has thus emerged as a critical application of AI in healthcare.

This project adopts a full-spectrum comparative approach, deliberately evaluating models across the entire complexity spectrum—from classical machine learning with handcrafted HOG features through to state-of-the-art Vision Transformers and Bayesian ensemble optimization. Rather than optimizing a single architecture, the goal is to produce a reproducible, fair benchmark providing both performance metrics and practical insights about what works best for medical image classification on a moderately sized, imbalanced dataset.

II. Dataset Description

In this work, we used a publicly available dataset from Kaggle titled *Chest X-Ray Images (Pneumonia)*. The dataset contains chest X-ray images labeled into two classes: *Normal* and *Pneumonia*. There are 5,863 X-Ray images (JPEG) and 2 categories (Pneumonia/Normal). The images are divided into three subsets: training, testing, and validation. These labeled images are used to train and evaluate the deep learning models for the pneumonia classification task.

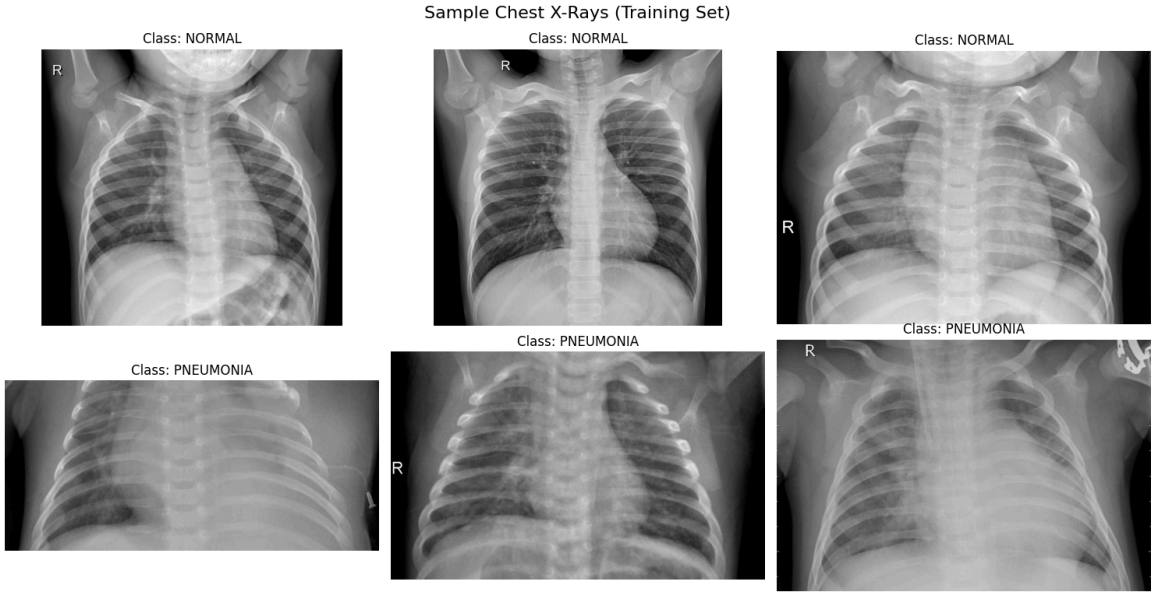

Figure 1. Sample chest X-ray images from the dataset showing Normal and Pneumonia cases

Table I. Dataset Distribution Across Splits and Classes

Split	NORMAL	PNEUMONIA	Total
Train	1,341	3,875	5,216
Validation	8	8	16
Test	234	390	624
Total	1,583	4,273	5,856

Figure 2a. Original dataset split

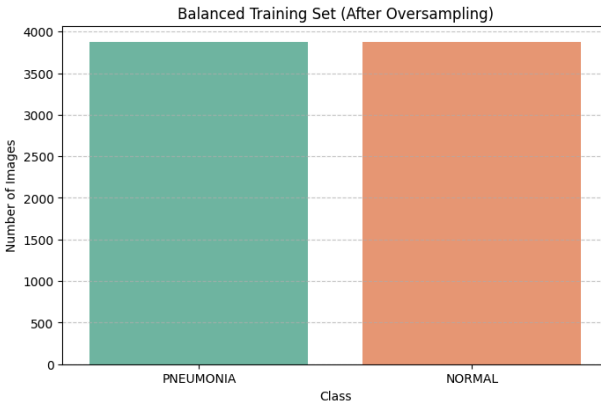

Figure 2b. Balanced training set after oversampling

Figure 2. Dataset distribution before and after oversampling.

Dataset Challenges and Preprocessing

Common benchmarks include ChestX-ray14 and the Kaggle pneumonia dataset (~5,800 images), with class imbalance addressed through augmentation and resampling strategies. Effective pneumonia detection requires comprehensive approaches addressing limited labeled samples, class imbalance, and imaging device variability. This project implements two complementary imbalance strategies: random oversampling for Keras-based models and random undersampling for PyTorch-based transformers. The preprocessing pipeline applies CLAHE and Gaussian Blur to all PyTorch models, directly addressing imaging variability. An augmentation pipeline of fifteen operations applied to VGG19 yields a +4.33% accuracy improvement, further validating the literature's emphasis on augmentation for smaller datasets.

Privacy-Preserving and Mobile Deployment

Ongoing work integrates federated learning for privacy-preserving pneumonia detection, enabling collaborative model training across multiple healthcare institutions without sharing sensitive patient data. This addresses critical regulatory concerns under HIPAA and GDPR, making AI-assisted detection viable for large-scale clinical deployment. Recent mobile-based systems implement explainable CNNs for field deployment in pediatric diagnosis, addressing model compression, inference speed, and point-of-care user-interface design. The lightweight MobileNetV2 evaluated here (7.5M parameters, 96.46% accuracy) is the most deployment-ready architecture, validating the literature's recommendation of efficient CNN architectures for real-world clinical contexts.

Project Objectives

III. Literature Review

Early CNN Approaches and Transfer Learning

Initial studies employed CNN models such as VGG-16 and VGG-19 for pneumonia classification from chest X-rays, reporting accuracies up to 95% on datasets such as ChestX-ray14. These models leveraged transfer learning to address the limited availability of medical imaging data, demonstrating superior performance compared to traditional ML in terms of sensitivity and specificity. Transfer learning has since become a cornerstone of pneumonia detection pipelines, with pre-trained architectures including DenseNet201 and ResNet50 providing strong feature representations. This project directly applies these findings: ResNet50 is used both as a frozen feature extractor and as a fine-tuned backbone, with fine-tuning on oversampled data improving test accuracy from 79.65% to 89.90%.

Systematic Reviews and Meta-Analyses

A 2023 systematic literature review analyzing over 50 studies identified CNNs, pre-trained models, and ensemble methods as top performers, while noting critical gaps in dataset diversity and model interpretability. Meta-analyses show that AI models achieve pooled sensitivity of 88% and specificity of 90% for pneumonia detection across approximately 12,000 chest X-rays, with ensemble approaches such as Random Forest and XGBoost reaching 92–96% accuracy using non-invasive biomarkers. These results motivate this project's inclusion of a classical HOG + Random Forest baseline (86.69%) as a direct comparator against deep learning models.

IV. Proposed System

The proposed system is a modular multi-model benchmarking pipeline for binary pneumonia classification from chest X-ray images, implemented across 12 Jupyter notebooks covering data preprocessing, individual model experiments, ablation studies, ensemble construction, and Bayesian hyperparameter optimization.

The key objectives are: (i) establish a classical ML baseline using HOG + Random Forest; (ii) design and train a custom CNN from scratch; (iii) explore unsupervised feature learning via an Autoencoder; (iv) evaluate the impact of CLAHE, augmentation, and class balancing; (v) benchmark Vision Transformers (Swin, DeiT) under controlled ablation conditions; (vi) build a heterogeneous four-model ensemble optimized with Optuna; (vii) provide clinical explainability via GradCAM; and (viii) produce a deployable ensemble configuration in JSON.

Development Environment

All experiments were executed on Kaggle GPU kernels using a Tesla T4 GPU (13,355 MB VRAM, Compute Capability 7.5). The software stack includes Python 3.12, PyTorch 2.10.0+cu128, TensorFlow (latest Kaggle build), timm 1.0.25, Optuna 4.0.0, grad-cam 1.5.4, scikit-learn, OpenCV, Pillow, and scikit-image.

Advanced Architectures and Attention Mechanisms

Recent models incorporating attention mechanisms and separable convolutions have achieved up to 98% accuracy in distinguishing bacterial, viral, and normal pneumonia cases. Transformer architectures combined with optimizers such as AdamW have further advanced pneumonia detection by capturing contextual, long-range dependencies in chest X-ray images. A CBAM-enhanced DenseNet121 with GradCAM explainability demonstrates that high diagnostic accuracy and clinical interpretability can be achieved simultaneously. This project applies analogous principles through Swin Transformer and DeiT experiments, employing AdamW with cosine annealing and GradCAM for the final Optuna-tuned ensemble.

Hybrid and Ensemble Models

Hybrid CNNs combining Xception and custom layers classify pneumonia with 96% accuracy, consistently outperforming single models through weighted ensembles. Transfer learning on VGG16 or ResNet50 with fine-tuning of final layers yields F1-scores exceeding 95% on augmented datasets. These findings directly inform this project's four-model ensemble strategy, which combines ResNet50, VGG19-BN, Swin-Base, and DeiT-Base via soft voting and Bayesian-optimized temperature scaling. The untuned soft-vote ensemble achieves near-perfect Pneumonia recall of 99.74%, confirming that heterogeneous ensembles outperform individual members on the clinically critical sensitivity dimension.

Data Processing Pipeline

A dedicated data processing notebook establishes the preprocessing pipeline shared by all Keras/TensorFlow-based models. Raw images are loaded using OpenCV and organized into a combined DataFrame with file paths and binary labels (0 = Normal, 1 = Pneumonia). Images are resized to 128×128×3 (RGB) tensors. Pixel normalization to [0,1] is applied, followed by an 80/20 stratified train-test split yielding 4,172 training and 1,044 test samples. TensorFlow's tf.data.Dataset API pipelines data with shuffling (buffer size 10,000) and batching (batch size 32), producing 194 training batches per epoch. Class imbalance is handled through random oversampling of the minority Normal class to 3,099 samples, matching the Pneumonia count, for a total of 6,198 balanced training samples.

Preprocessing for PyTorch Models

For all PyTorch-based models a richer preprocessing pipeline is implemented via a custom XRayDataset class applying CLAHE on-the-fly during __getitem__: CLAHE is applied on the L-channel (LAB color space) to enhance local contrast of pneumonia-related lung opacities; Gaussian Blur (3×3, σ=0.5) provides mild denoising; images are resized to 256 then center-cropped to 224×224; grayscale-to-RGB conversion replicates the L-channel across 3 channels to satisfy pretrained model input requirements; and ImageNet normalization is applied. The augmented pipeline additionally applies RandomCrop with reflect padding, RandomHorizontalFlip, RandomVerticalFlip, RandomRotation (±10°), ColorJitter (brightness/contrast 0.15), RandomAdjustSharpness, RandomAutocontrast, and RandomErasing (p=0.25).

Class Imbalance: Two Strategies

Two strategies are employed depending on framework. Oversampling (Keras models) duplicates the minority Normal class until it matches the Pneumonia count (3,099 each), yielding 6,198 balanced samples. Random Undersampling (PyTorch models) downsamples the majority Pneumonia class (3,875 samples) to match the Normal count (1,341), producing 2,682 balanced samples. This reduces memorization risk and halves epoch duration (~100 s vs. ~200 s), at the cost of discarding 2,534 majority-class samples.

V. Model Architectures

HOG + Random Forest

HOG features are extracted from each 128×128×3 training image using scikit-image's hog() with default parameters. The resulting fixed-length feature vectors train a Random Forest classifier (1,000 estimators, random_state=42) on the raw imbalanced dataset. This model serves as the classical ML baseline.

Custom CNN

A custom Sequential CNN is designed from scratch with TensorFlow/Keras: Conv2D(32, 3×3, same) + ReLU + MaxPool; Conv2D(64, 3×3, same) + ReLU + MaxPool; Flatten + Dense(128, ReLU) + Dense(1, Sigmoid). Total parameters: 939,041 (3.58 MB). Trained with Adam optimizer and binary cross-entropy for 10 epochs on the oversampled balanced dataset.

Autoencoder-based Classifier

Stage 1: A convolutional autoencoder (21,283 parameters) is pre-trained for 5 epochs to reconstruct 128×128×3 inputs, compressing to a 16×16×32 latent space (MSE loss reaching 0.0019 on epoch 5). Stage 2: The frozen encoder produces 8,192-dimensional feature vectors that train a Dense classifier (1,048,833 parameters) for 5 epochs, achieving 93.10% training accuracy by epoch 5.

MobileNetV2

MobileNetV2 (128×128 input variant, 2,257,984 frozen backbone parameters) is fine-tuned with a custom head: GlobalAveragePooling2D → Dense(128, ReLU) → Dense(1, Sigmoid). Total parameters: 7,501,377 (28.62 MB). Trained on 6,198 balanced samples for 10 epochs. Training accuracy reaches 99.39% by epoch 9; validation accuracy exceeds 96% throughout.

ResNet50

Two configurations are evaluated. Configuration 1 (Frozen): ResNet50 with ImageNet weights, frozen backbone (23,587,712 parameters) plus a Dense head (2,049 trainable parameters), trained on 5,216 images without oversampling. Test accuracy: 79.65% (Precision: 0.7563, Recall: 0.9949, F1: 0.8594). Configuration 2 (Fine-tuned): Same architecture trained on oversampled balanced data (7,750 samples) for up to 30 epochs with early stopping and augmentation. Test accuracy: 89.90% (Precision: 0.9077, Recall: 0.9333, F1: 0.9204).

EfficientNetB0

EfficientNetB0 (16.7 MB) is used as a frozen feature extractor with only 1,281 trainable parameters (head: GlobalAveragePooling2D → Dropout(0.3) → Dense(1, Sigmoid)). Input: 224×224. Trained on 7,750 balanced images for up to 30 epochs with early stopping. Test accuracy: 89.42% (Precision: 0.8971, Recall: 0.9385, F1: 0.9173).

VGG19 Fine-Tuning

VGG19 (pretrained, 126.99M parameters) is fine-tuned in two variants using the downsampled balanced dataset (2,682 samples). Anti-overfitting strategy: backbone frozen for 3 epochs (head lr=10^{-4}), then progressive unfreeze (backbone lr=10^{-5}), AdamW (weight_decay=10^{-3}), CrossEntropyLoss (label_smoothing=0.1), Cosine Annealing, Early Stopping

(patience=5). Variant 1 (No Augmentation): Test accuracy 86.70%, F1 0.8643, training time 581 s. Variant 2 (With 15-step Augmentation): Test accuracy 91.03%, F1 0.9088, training time 1327 s. Augmentation improved accuracy by +4.33%.

Swin Transformer — Initial Study

swin_base_patch4_window7_224 (86.75M parameters) from timm is trained with Adam (lr=10^{-4}) on the full training set for up to 10 epochs. Training accuracy reaches 99.52% by epoch 5 while validation accuracy collapses to 77.56%—a critical overfitting problem motivating the rigorous anti-overfitting framework used in all subsequent PyTorch experiments. Swin Baseline (No Aug): Test accuracy 92.15%, F1 0.9207. Swin Augmented: Test accuracy 91.03%, F1 0.9080.

Swin Transformer — SE Block Ablation

A 4-variant ablation study isolates the effects of SE block and data augmentation independently. All variants use the improved framework: random downsampling (2,682 balanced samples), progressive unfreeze (3 frozen epochs, then lr/10), AdamW (weight_decay=10^{-3}), CrossEntropyLoss (label_smoothing=0.1), Dropout(0.4), Cosine Annealing LR, and Early Stopping (patience=5). The SE block is implemented as spatial feature recalibration on the 1,024-channel spatial feature map (7×7) from Swin's last stage. SE adds 525,568 parameters.

Table II. Swin Transformer Ablation Study Results

Var.	SE / Preprocessing	Acc.	F1	Time (s)
A (Best)	No SE + Baseline	91.83%	0.9166	1,039
B	SE + Baseline	89.90%	0.8963	1,034
C	No SE + Augmented	91.51%	0.9138	1,697
D	SE + Augmented	90.87%	0.9069	1,701

DeiT — Three-Setup Study

DeiT-Base (deit_base_patch16_224, 86.19M parameters) is evaluated in three configurations using the same anti-overfitting framework. DeiT uses 16×16 patches (196 patches per image) with global self-attention, a CLS token, and a Distillation token. The SE block for DeiT operates on the 1D CLS token (768-dim) rather than spatial features (SE adds 295,872 parameters).

Four-Model Ensemble

The ensemble notebook trains four distinct PyTorch backbones from scratch using a unified configuration: img_size=224, batch_size=16, epochs_freeze=2, epochs_finetune=5, lr_freeze=3×10^{-4}, lr_finetune=3×10^{-5}, weight_decay=10^{-4}, label_smoothing=0.1, Dropout=0.4, SE_block=True. The downsampled balanced set (2,682 total) is split into train=2,413, val=269, test=624.

Three ensembling strategies are evaluated using saved softmax probability arrays. Soft Voting takes the mean of softmax probabilities. Weighted Soft Voting weights by validation accuracy (ResNet50=0.248, VGG19=0.252, Swin=0.251, DeiT=0.249—nearly uniform). Hard Voting uses majority argmax; ties broken by soft probability. All three ensemble strategies achieve near-perfect recall (99.74%) for the Pneumonia class.

Optuna Bayesian Hyperparameter Optimization

Starting from the four saved checkpoints, a two-phase Bayesian optimization pipeline is constructed using Optuna 4.0.0 with TPESampler (multivariate=True, seed=42). Phase 1 (200 trials): The 9-dimensional search space covers per-model temperatures Ti ∈ [0.5, 3.0], per-model weights wi ∈ [0.01, 1.0] (normalized to sum=1), and decision threshold ∈ [0.10, 0.90]. Baseline untuned ensemble: val acc=96.55%, test acc=91.19%, test F1=0.9337.

Phase 1 best configuration: ResNet50 T=1.893, VGG19 T=2.960, Swin T=2.049, DeiT T=2.069, threshold=0.252. Phase 2 (15 trials): The weakest Phase 1 member (VGG19, weight=0.128) is identified for retraining. Optuna searches over five training hyperparameters using HyperbandPruner. VGG19 is retrained for 11 epochs, achieving val_acc=0.9866 (Δ=+0.023 over original). Final ensemble configuration: ResNet50 T=1.439, weight=0.120; VGG19 T=0.828, weight=0.413; Swin T=1.824, weight=0.187; DeiT T=1.695, weight=0.280; threshold=0.158.

Table III. DeiT Three-Setup Comparison

Setup	Acc.	F1	Time (s)
Baseline (No Aug)	89.90%	0.8961	1,041
Augmented	90.87%	0.9072	1,698

SE + Augmented	90.38%	0.9011	1,695

Table IV. Individual Ensemble Members — Validation and Test Results

Model	Val Acc	Test Acc	Recall	F1
ResNet50	0.9665	0.9135	0.9641	0.9330
VGG19-BN	0.9814	0.8926	0.9872	0.9200
Swin-Base	0.9777	0.8734	0.9974	0.9078
DeiT-Base	0.9703	0.8702	0.9974	0.9057

VI. Results

Comprehensive Performance Summary

All models are evaluated on the same held-out test set of 624 images (234 Normal, 390 Pneumonia). Table V provides a complete performance summary across all models and configurations.

Table V. Complete Model Performance on Test Set (624 Images)

Model / Config	Acc.	Prec.	Rec.	F1
HOG + RF (1000 trees)	86.69%	0.87	0.87	0.86
Custom CNN (scratch)	92.43%	0.93	0.92	0.93
Autoencoder + Classifier	88.41%	0.91	0.88	0.89
MobileNetV2 (fine-tuned)	96.46%	0.97	0.96	0.96
ResNet50 (frozen, no bal.)	79.65%	0.756	0.995	0.859
ResNet50 (fine-tuned, OS)	89.90%	0.908	0.933	0.920
EfficientNetB0 (frozen)	89.42%	0.897	0.939	0.917
VGG19 (No Aug)	86.70%	0.868	0.867	0.864
VGG19 (With Aug)	91.03%	0.912	0.910	0.909
Swin-Base Baseline (full)	92.15%	0.922	0.921	0.921
Swin-Base Aug (full)	91.03%	0.916	0.910	0.908
Swin-A: No SE + Baseline	91.83%	0.922	0.918	0.917
Swin-B: SE + Baseline	89.90%	0.905	0.899	0.896
Swin-C: No SE + Aug	91.51%	0.917	0.915	0.914
Swin-D: SE + Aug	90.87%	0.912	0.909	0.907
DeiT-Base Baseline	89.90%	0.906	0.899	0.896
DeiT-Base Augmented	90.87%	0.911	0.909	0.907
DeiT + SE + Aug	90.38%	0.911	0.904	0.901
Ensemble Soft Vote	89.58%	0.859	0.997	0.923
Ensemble Weighted Soft	89.58%	0.859	0.997	0.923
Ensemble Hard Vote	89.26%	0.855	0.997	0.921
Baseline Soft Vote (TTA)	91.19%	0.882	0.992	0.934

Confusion Matrices

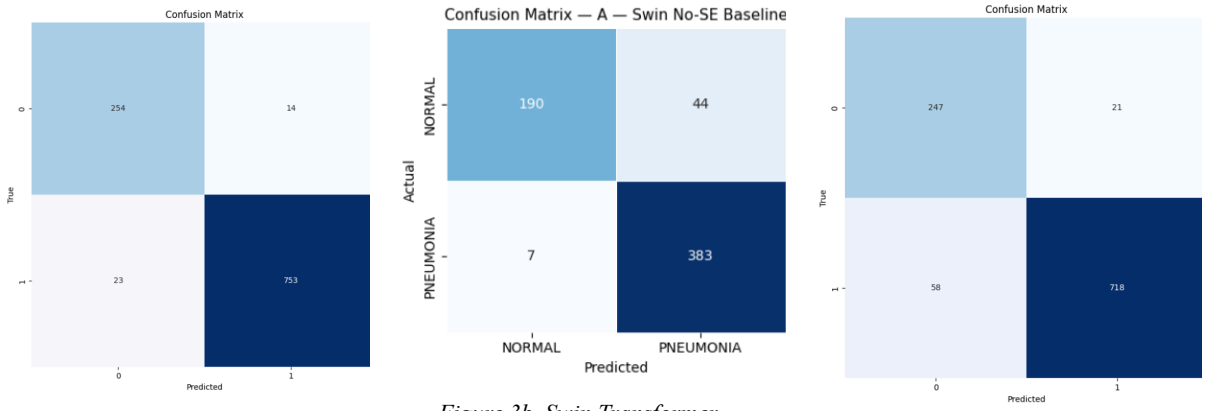

Figure 3a. MobileNetV2 *Figure 3b. Swin Transformer* *Figure 3c. Custom CNN*

Figure 3. Confusion matrices for different models.

Ablation Study Insights

The Swin Transformer ablation across 4 variants in a 2×2 factorial design reveals counter-intuitive results attributable to the small balanced training set (2,682 samples): the SE block consistently hurts performance (−1.93% on baseline, −0.64% on augmented); augmentation alone very slightly reduces accuracy (−0.32% without SE), and only provides a minor benefit when combined with SE (+0.97%). DeiT shows the same pattern: augmentation helps (+0.97%) but SE hurts (−0.49%), confirming this is a dataset-size effect rather than architecture-specific behavior. The Swin's shifted-window attention already provides implicit feature reweighting, making additional SE-style recalibration redundant or counterproductive.

Per-Class Analysis

The ensemble achieves the highest Pneumonia recall (99.7%) at the cost of Normal recall (72.7%), reflecting the clinical priority of maximizing sensitivity. MobileNetV2 achieves the best overall balance across both classes.

Table VI. Per-Class Metrics for Key Models

Model	N-Pr	N-Re	P-Pr	P-Re
HOG + RF	0.88	0.56	0.86	0.97
CNN	0.81	0.92	0.97	0.93
Autoencoder	0.70	0.97	0.99	0.86
MobileNetV2	0.92	0.95	0.98	0.97
VGG19 (Aug)	0.94	0.82	0.90	0.97
Swin-A (Best)	0.96	0.81	0.90	0.98
DeiT Aug	0.94	0.81	0.90	0.97
Ensemble Soft	0.994	0.727	0.859	0.997

N-Pr = Normal Precision, N-Re = Normal Recall, P-Pr = Pneumonia Precision, P-Re = Pneumonia Recall

Final Optuna Ensemble Configuration

The configuration reflects the Bayesian optimization result: VGG19 dominates the ensemble (weight=0.413) after Phase 2 retraining, with a sharpened temperature (T=0.828). The very low decision threshold (0.158) reflects medical prioritization of recall—the model flags Pneumonia if ≥15.8% of ensemble probability mass lies on the Pneumonia class.

Explainable AI Visualization

GradCAM visualizations confirm the ensemble focuses on bilateral lower lobe regions in Pneumonia cases and clear lung fields in Normal cases, consistent with radiological findings. The heatmaps highlight regions that contribute most to the model's decision. For Pneumonia cases, the model consistently attends to localized high-intensity regions in the lower lung zones, aligning with clinically relevant opacities. In contrast, Normal cases show diffuse or minimal activation across clear lung fields. Misclassified examples reveal either weak or spatially ambiguous activations, suggesting borderline patterns or dataset noise. The weighted ensemble further

stabilizes attention by aggregating model-specific saliency maps, producing more consistent and anatomically meaningful focus areas.

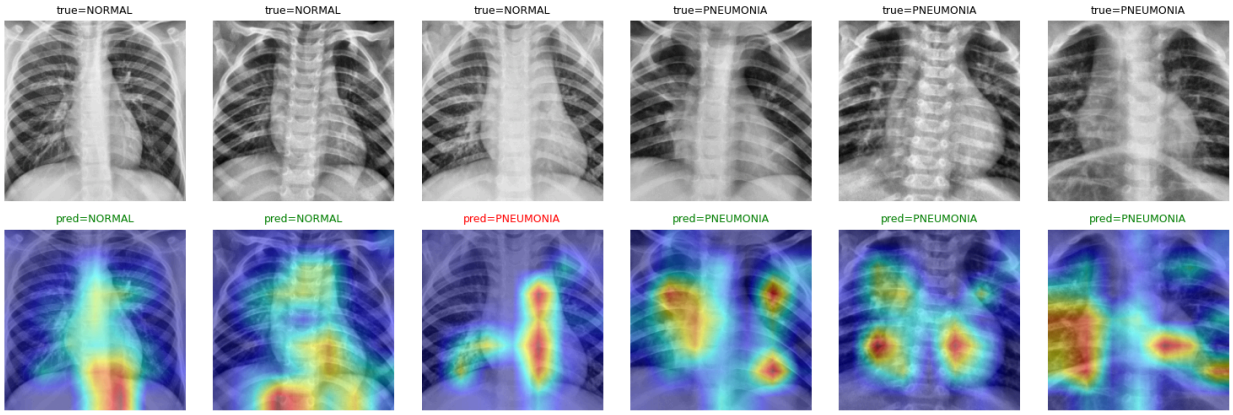

Figure 4. Grad-CAM based Explainable AI visualization of the optimized ensemble. The figure shows true vs predicted labels alongside corresponding heatmaps. The model focuses on clinically relevant lung regions, particularly the lower lobes in Pneumonia cases, while maintaining diffuse attention in Normal cases.

VII. Conclusion

This project delivers a rigorous, reproducible, and comprehensive evaluation of the full spectrum of machine learning approaches for automated pneumonia classification from chest X-ray images. Beginning with handcrafted HOG features and a Random Forest baseline (86.69%), the study progressively advances through custom CNNs (92.43%), autoencoder-based feature learning (88.41%), transfer learning with MobileNetV2 (96.46%) and ResNet50 (89.90%), EfficientNetB0 (89.42%), VGG19 fine-tuning (91.03%), Vision Transformers in controlled ablation studies (Swin: up to 92.15%, DeiT: up to 90.87%), and finally a four-model ensemble with Bayesian hyperparameter optimization achieving near-perfect Pneumonia recall (99.74%).

The key finding is that pretrained transfer learning, particularly MobileNetV2—provides the best single-model accuracy for this dataset size and domain. Vision Transformers are competitive but require substantially more regularization effort and compute, and their additional complexity (87M+ parameters) is not justified by the accuracy gains over much smaller pretrained CNNs. The ensemble excels in the clinically critical dimension of sensitivity, making it the preferred choice for a screening deployment.

Future directions include: (i) semi-supervised learning to leverage large repositories of unlabeled chest X-rays; (ii) multi-class classification distinguishing bacterial vs. viral pneumonia; (iii) integration with clinical metadata (patient age, symptoms, laboratory results) for multimodal diagnosis; (iv) knowledge distillation from the ensemble into a lightweight student model for edge deployment; and (v) prospective clinical validation on an independent hospital dataset.

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